1: Python Basic

1. Variables and Data Types:

Variables store information.

```
name = "Rajaram"
age = 19
cgpa = 8.5
```

Data Types:

- String (str)
- Integer (int)
- Float (float)
- Boolean (True/False)

Example:

```
student = "Raj"
marks = 90
percentage = 85.5
passed = True
```

Why Needed?

Student names, marks, attendance, and course details are stored using variables.

2. Input and Output:

```
name = input("Enter Name: ")
print("Welcome", name)
```

Why Needed?

To take user inputs in small programs.

3. Operators:

Arithmetic Operators:

```
a = 10
b = 5

print(a+b)
print(a-b)
print(a*b)
print(a/b)
```

Comparison Operators

```
print(a > b)
print(a == b)
```

Why Needed?

To calculate averages, percentages, and compare student performance.

4. Conditional Statements:

If-Else:

```
marks = 85

if marks >= 40:
    print("Pass")
else:
    print("Fail")
```

Why Needed?

To identify:

- Passed students
- Failed students
- Low performers

5. Loops:

For Loop:

```
for i in range(5):
    print(i)
```

While Loop:

```
count = 1

while count <= 5:
    print(count)
    count += 1
```

Why Needed?

Suppose there are 100 students.Instead of writing:

```
print(student1)
print(student2)
```

6. Functions:

Functions reduce repeated code.

```
def average(a, b):
    return (a+b)/2

print(average(80,90))
```

Why Needed?

You may calculate attendance percentage many times.

Create one function and reuse it.

2: Data Structures

7. Lists:

Most important for beginners.

```
marks = [75,80,90,85]

print(marks)
print(marks[0])
```

Why Needed?

Store:

- Marks
- Attendance
- Scores

Example:

```
attendance = [90,85,88,92]
```

8. Dictionaries:

Store data as key-value pairs.

```
student = {
    "Name":"Raj",
    "Marks":90,
    "Attendance":85
}

print(student["Name"])
```

Why Needed?

Represent a student's complete record.

9. Tuples:

```
data = (101, "Raj", 85)
```

10. Sets:

```
subjects = {"Python","AI","ML"}
```

Useful for removing duplicates.

3: File Handling

Read CSV File:

```
file = open("data.txt","r")

print(file.read())

file.close()
```

Why Needed?

Student data is usually stored in:

- CSV files
- Excel files

4: Exception Handling

```
try:
   x = 10/0
except ZeroDivisionError:
   print("Cannot divide by zero")
```

Why Needed?

Prevents program crashes.

5: Object-Oriented Programming

```
class Student:

    def __init__(self,name,marks):
        self.name = name
        self.marks = marks

s1 = Student("Raj",90)

print(s1.name)
```

Why Needed?

Can represent students, courses, and reports as objects.

6: Libraries Used

1. NumPy:

Used for numerical calculations.

```
import numpy as np

marks = np.array([80,90,70,85])

print(np.mean(marks))
```

Use Cases:

- Average marks
- Highest marks
- Statistical calculations

2. Pandas:

```
import pandas as pd

data = pd.read_csv("students.csv")

print(data.head())
```

Use Cases:

- Read datasets
- Clean data
- Filter students
- Analyze performance

3. Matplotlib

```
import matplotlib.pyplot as plt
marks = [70,80,90,85]
plt.plot(marks)
plt.show()
```

Use Cases:

- Line graphs
- Bar charts
- Attendance trends